

sxSNF: A Novel Method for Single-Cell Multi-Modal Data Integration by Combining Similarity Network Fusion and Deep Graph Learning

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Abstract

The precise characterization of cell types is critical for deciphering cellular heterogeneity. Although single-cell sequencing technologies enabled simultaneous measurement of multiple molecular modalities, integrating these diverse data types to achieve robust clustering remains a significant challenge. To address this, we present a novel framework that combines similarity network fusion (SNF) with deep graph learning. Our method iteratively calculates and fuses similarity networks from distinct omics profiles, thereby constructing a more robust consensus graph that overcomes the limitations of conventional graph construction for multimodal data. This integrated graph effectively preserves complementary local features from different modalities while mitigating technical batch effects. Comprehensive benchmarks demonstrate that our SNF-enhanced graph model outperforms existing state-of-the-art methods in clustering accuracy. The code for sxSNF is available at <https://github.com/labxscut/sxSNF>.

Keywords

single-cell multimodal integration; similarity fusion network; cell clustering

ACM Reference Format:

Hongyu Duan, Yang Wang, Qianwen Chen, Xiaoxia Zhen, Jianxiu Jin, Huiling Liu, and Li C. Xia. 2018. sxSNF: A Novel Method for Single-Cell Multi-Modal Data Integration by Combining Similarity Network Fusion and Deep Graph Learning. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Integrating multi-modal data in single-cell studies has become essential for advancing our understanding of cellular heterogeneity and complex biological systems [Wang et al. 2021]. Multi-modal data, such as scRNA-seq (gene expression), scATAC-seq (chromatin accessibility), and CITE-seq (protein abundance), offer complementary perspectives of the same cell, providing a more complete and holistic view of cellular processes [Satija et al. 2015; Wolf et al. 2018]. For example, scRNA-seq reveals gene expression levels, scATAC-seq provides insights into chromatin accessibility that regulates gene expression, and CITE-seq captures protein abundance on the cell surface. By integrating these multiple data layers, we can uncover a deeper understanding of cell differentiation, disease mechanisms, and therapeutic responses [Cortal et al. 2021]. However, the challenge lies in effectively combining these different data types to

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Conference acronym 'XX, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/2018/06
<https://doi.org/XXXXXXX.XXXXXXX>

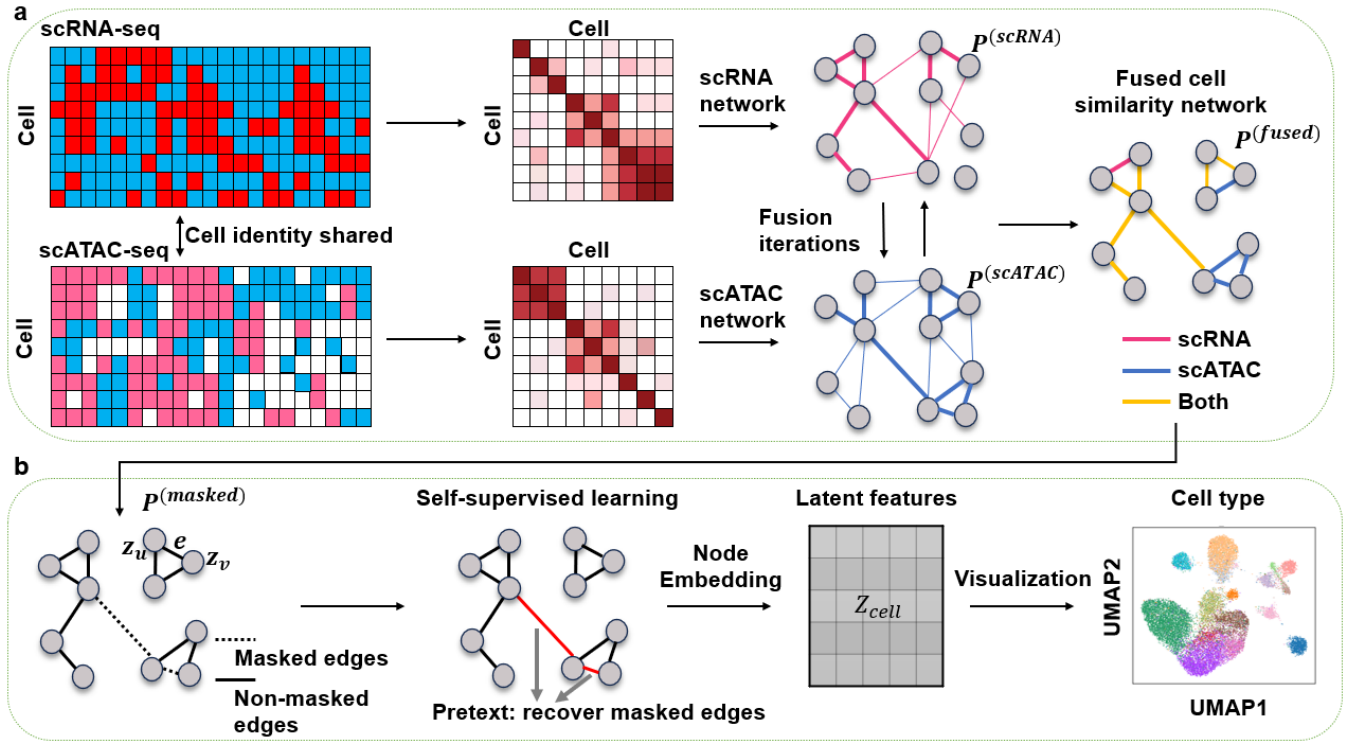


Figure 1: Schematic of the sxSNF framework. (a) Multi-modal Graph Construction and Fusion: creates cell similarity graphs from scRNA-seq and scATAC-seq data, followed by an iterative fusion process to align them. (b) Unified Representation Learning and Downstream Analysis: transform fused network into a deep cell embedding for latent feature extraction, and cluster visualization.

maximize their biological insights, especially considering the heterogeneity, sparsity, and scale of single-cell data [Welch et al. 2019].

In recent years, several computational methods have been developed to tackle the problem of multi-modal data integration. Statistical techniques, such as Canonical Correlation Analysis (CCA) and Multiple Correspondence Analysis (MCA), align features from different modalities by finding linear relationships between them [Stuart et al. 2019]. For example, Seurat’s Weighted Nearest Neighbor (WNN) analysis incorporates CCA to balance shared and modality-specific signals, effectively integrating RNA-seq and protein data [Ma et al. 2020]. Matrix factorization methods, such as MOFA+ (Multi-Omics Factor Analysis), decompose data matrices into shared and unique components to uncover underlying factors driving multi-modal variation [Hao et al. 2021]. However, they are limited by their inability to capture complex non-linear relationships, which are prevalent in single-cell data.

Machine learning-based approaches address these limitations by leveraging non-linear transformations to map data into shared latent spaces [Korsunsky et al. 2019]. For instance, Deep Canonical Correlation Analysis (DeepCCA) [Stuart et al. 2019] extends CCA by using deep neural networks to learn more complex relationships between modalities [Bakken et al. 2021]. Autoencoders, such as those implemented in SIMBA and scVI [Mimitou et al. 2021], compress high-dimensional data into lower-dimensional embeddings, enabling efficient integration and clustering [Heumos et al. 2023].

Generative models like totalVI [Argelaguet et al. 2021] further improve integration by probabilistically reconstructing missing data and accounting for technical noise. However, these methods can suffer from computational inefficiency, particularly with large-scale datasets, and are often prone to overfitting in cases of sparsity and batch effects. Recent frameworks, such as scJoint [Lin et al. 2022] and MultiVI [Lin et al. 2022], have attempted to address these issues by integrating batch correction and feature alignment directly into their architectures.

To address these issues, we propose sxSNF, a novel integration method that combines Similarity Network Fusion (SNF) [Gong et al. 2021] with Graph Neural Networks (GNNs) [Argelaguet et al. 2020; Ciortan and Defrance 2022]. sxSNF leverages the iterative refinement of similarity networks through SNF, capturing relationships across multiple modalities and reducing batch effects [Ashuach et al. 2023]. After fusion, the integrated data is processed using GNNs [Lin et al. 2022], which use graph convolutional layers to learn robust [Kang et al. 2021], low-dimensional embeddings that preserve both local and global structures [Cao and Gao 2022]. GNNs are well-suited to handle complex, non-linear relationships, making sxSNF particularly effective for multi-modal data where interactions across molecular layers are intricate and dynamic [Song et al. 2021; Wen et al. 2022]. The use of GNNs also ensures that the integrated data can scale efficiently [Song et al. 2021], addressing

Table 1: Dataset Adopted in This Study

Dataset	Cells	Genes (scRNA)	Peaks (scATAC)
PBMC-10k	11909	36601	108377
Chen-2019	9190	28930	241757
SHARE-seq	34774	17000	370542

computational concerns when processing large datasets. This hybrid approach can overcome key limitations of existing methods, improving clustering accuracy, robustness against noise and sparsity, and better preservation of modality-specific information [Shao et al. 2021].

We organize this paper as follows: Section 1 introduces the problem of single-cell multimodal data integration, highlighting the challenges of capturing cellular heterogeneity and the limitations of existing methods. Section 2 presents sxSNF, describing its algorithm and how the integration of SNF with GNNs addresses the challenges identified in previous methods. Section 3 evaluates sxSNF through experimental results, demonstrating significant performance improvements, showcasing its superiority in handling multi-modal data. Finally, Section 4 discusses the implications of sxSNF for future single-cell research and its potential applications in other areas of data integration.

2 Methods

2.1 Overview of sxSNF framework

In Figure 1, we outline the two-stage sxSNF framework for single-cell multi-omics integration. In Figure 1a, gene expression (scRNA) and chromatin accessibility (scATAC) data are preprocessed separately, generating modality-specific similarity maps. These are converted into k-nearest neighbor graphs (pink for scRNA, blue for scATAC), where edge weights reflect distance. An iterative fusion algorithm aligns the graphs by transferring edge weights and reinforcing edges supported by both modalities to produce a unified network. The fused network undergoes self-supervised learning to produce a low-dimensional embedding (Figure 1b). The core of this learning is a *pretext task* of reconstructing masked edges in the graph. By masking and reconstructing true edges via a contrastive loss, the model learns denoised, structure-preserving representations. Uniform Manifold Approximation and Projection (UMAP) [McInnes et al. 2018] visualization was employed, revealing distinct cell-type clusters. This workflow enables robust integration, visualization, and analysis of multi-modal single-cell data.

2.2 Datasets and comparative methods

We evaluate sxSNF and other baseline methods on three multimodal single-cell datasets: PBMC-10k (human peripheral blood; 10x Multiome) [Yang et al. 2016], Chen-2019 (mouse/human brain; SNARE-seq) [Chen et al. 2019], and SHARE-seq (mouse skin; SHARE-seq) [Cortal et al. 2021]. Their basic statistics are detailed in Table 1.

We compare sxSNF with spectral clustering model [Von Luxburg 2007], and state-of-art methods including the SIMBA model [Chen et al. 2023], the scMIC model [Zhan et al. 2023], Guanlab-Dengkw method [Rautenstrauch and Ohler 2024] and the DCCA method

[Zebende 2011], and the scMIC+SNF method, in which the scMIC method is augmented by SNF.

2.3 Single-cell data preprocessing

For scRNA-seq data, preprocessing was performed using Scanpy [Wolf et al. 2018]. Genes detected in fewer than 10 cells were filtered out [Wolf et al. 2018]. Raw counts were normalized per cell and log-transformed. Highly variable genes (mean: 0.0125, dispersion: > 0.5) were selected to retain 2,000 genes for downstream analysis.

For scATAC-seq data, peaks found in fewer than 100 cells were removed. Variable peaks were identified by performing singular value decomposition (SVD) on the binarized matrix. The top components were selected via the scree plot elbow, and peaks with high absolute loadings (top 5%) across these components were used as final variable peak features [Cusanovich et al. 2018].

2.4 Similarity Network Fusion

Our similarity network fusion (SNF) approach constructs a network for each modal data, and then based on the principle of "neighbor information sharing", the constructed single modal network are efficiently fused into a full-spectrum network. The fused network is capable of capturing both the shared and complementary information from various data modalities, which mitigates the impact of modalities noise and batch effects.

2.4.1 Similarity network for each modality.

For each modality of the n cells, we formulate the a graph $G = (V, E)$, where the vertex set $V = \{v_1, v_2, \dots, v_n\}$ represents cells, and edges E are weighted by pairwise cell similarities. These weights are encoded in a symmetric similarity matrix $W \in \mathbb{R}^{n \times n}$, with $W(i, j)$ defined using a scaled exponential kernel:

$$W(i, j) = \exp \left(-\frac{\rho^2(v_i, v_j)}{\mu \cdot \epsilon_{i,j}} \right) \quad (1)$$

Here, $\rho(v_i, v_j)$ denotes the distance between cells v_i and v_j , and $\mu \in [0.3, 0.8]$ is a tunable hyperparameter. The scaling factor $\epsilon_{i,j}$ mitigates local density variations and is computed as:

$$\epsilon_{i,j} = \frac{\frac{1}{n} \sum_{i=1}^n \rho(v_i, N_i) + \frac{1}{n} \sum_{j=1}^n \rho(v_j, N_j) + \rho(v_i, v_j)}{3} \quad (2)$$

Let N_i denote the neighbor set of cell v_i obtained via K-nearest neighbors (KNN), where $\rho(v_i, N_i)$ represents the average distance from cell v_i to its neighbors.

To enhance numerical stability during normalization, we avoid standard row-normalization, which can be sensitive to diagonal dominance. Instead, we apply the following robust normalization:

$$P(i, j) = \begin{cases} \frac{W(i, j)}{2 \sum_{k \neq i} W(i, k)}, & j \neq i \\ 0.5, & j = i \end{cases} \quad (3)$$

We further refine the local affinity structure using the sparse requirement:

$$S(i, j) = \begin{cases} \frac{W(i, j)}{\sum_{k \in N_i} W(i, k)}, & j \in N_i \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Table 2: Benchmark results of seven single-cell multimodal integration methods on the PBMC-10k dataset.

Methods	sxSNF	SIMBA	SNF+scMIC	scMIC	DCCA	Guanlab-dengkw	Spectral clustering
ARI	0.5584	0.4854	0.3523	0.1142	0.3375	0.2683	0.0991
NMI	0.7274	0.6839	0.5732	0.2841	0.5798	0.5164	0.3633
AMI	0.7260	0.6822	0.5710	0.2802	0.5777	0.5140	0.3608

Table 3: Benchmark results of seven single-cell multimodal integration methods on SHARE-seq dataset.

Methods	sxSNF	SIMBA	SNF+scMIC	scMIC	DCCA	Guanlab-dengkw	Spectral clustering
ARI	0.589	0.512	0.421	0.153	0.394	0.325	0.127
NMI	0.768	0.724	0.634	0.342	0.621	0.567	0.398
AMI	0.742	0.701	0.631	0.327	0.602	0.519	0.397

This construction ensures that similarities between non-adjacent cells are explicitly set to zero.

In our study, we adopt the above framework for the raw modality graph embedding. We also investigate the impact of distance metrics:

- Cosine Distance. We test ρ with cosine distance. Cosine distance focuses on the directional agreement between data points. This makes it more effective for high-dimensional data, where the overall pattern (direction) is more important than the magnitude of individual features.
- Chi-Square Distance. For discrete or binned data, such as single-cell RNA sequencing count matrices, we test chi-square distance. This metric is particularly suited for frequency data, as it measures the discrepancy between two distributions while accounting for their expected proportions.

2.4.2 Similarity at modality networks fusion. The similarity network for each modality is characterized by two matrices: the global state matrix P (Eq. 3), which captures full pairwise similarities between all cells, and the sparse kernel matrix S (Eq. 4), which encodes only the local similarities to the k -nearest neighbors. In the next fusion process, P serves as the initial state, while S acts as the diffusion kernel, effectively preserving local graph structure and enhancing computational efficiency.

Following the preceding graph construction steps, we obtain similarity matrices $W^{(v)}$, state matrices $P^{(v)}$, and kernel matrices $S^{(v)}$ for each modality $v = \{\text{scRNA}, \text{scATAC}\}$. The core of Similarity Network Fusion (SNF) lies in the iterative cross-diffusion of state matrices:

$$P_{t+1}^{(\text{scRNA})} = S^{(\text{scRNA})} \cdot P_t^{(\text{scATAC})} \cdot \left(S^{(\text{scRNA})}\right)^T \quad (5)$$

$$P_{t+1}^{(\text{scATAC})} = S^{(\text{scATAC})} \cdot P_t^{(\text{scRNA})} \cdot \left(S^{(\text{scATAC})}\right)^T \quad (6)$$

Here, $P_{t+1}^{(\text{scRNA})}$ and $P_{t+1}^{(\text{scATAC})}$ denote the updated state matrices after t iterations for scRNA and scATAC modalities, respectively. This update scheme creates two mutually reinforcing diffusion processes. After t iterations, the consensus matrix $P^{(\text{fused})}$ (see Figure 1a) is computed as:

$$P^{(\text{fused})} = \frac{P_{t+1}^{(\text{scRNA})} + P_{t+1}^{(\text{scATAC})}}{2} \quad (7)$$

Without loss of generalizability, the update rule of $P^{(\text{scRNA})}$ can be expressed at the element level as:

$$P_{t+1}^{(\text{scRNA})}(i, j) = \sum_{k \in N_i} \sum_{l \in N_j} S^{(\text{scRNA})}(i, k) \cdot S^{(\text{scRNA})}(j, l) \cdot P_t^{(\text{scATAC})}(k, l) \quad (8)$$

This formulation reveals the intuition behind SNF: if two cells v_i and v_j share common neighbors across both modalities, they are more likely to belong to the same cluster. The process allows similarity information to propagate across modalities—even if two cells are not directly similar in one modality, their similarity can be inferred through the structural context provided by the other ($P^{(\text{scATAC})}$ updates similarly).

The algorithm naturally generalizes to $m > 2$ modalities. The iterative update rule becomes:

$$P^{(v)} = S^{(v)} \cdot \left(\frac{\sum_{k \neq v} P^{(k)}}{m-1} \right) \cdot \left(S^{(v)}\right)^T, \quad v = 1, 2, \dots, m \quad (9)$$

Through this diffusion process, the SNF effectively suppresses weak edges (potential noise) while amplifying strong, consistent similarities present across modalities. This enables the method to fully leverage the local structure of each network and integrate both shared and complementary information, resulting in a robust consensus representation.

2.5 Deep Graph Learning

Based on the fused similarity matrix $P^{(\text{fused})}$, we employ a self-supervised learning approach to reconstruct a multimodal graph and learn unified low-dimensional embeddings that capture robust cell-cell similarities. The core idea involves randomly masking a subset of edges in the fused graph and training the model to recover the masked connectivity patterns.

Let $Z \in \mathbb{R}^{n \times d}$ denote the final embedding matrix, where $\mathbf{z}_i \in \mathbb{R}^d$ is the representation of the i -th cell. The affinity between any two connected cells u and v is modeled as $s_{uv} = \mathbf{z}_u^\top \mathbf{z}_v$.

To learn these embeddings, we formulate the task as edge reconstruction through masking and recovery. Specifically, we create a masked version of the fused graph $P^{(\text{masked})}$ by randomly selecting a subset of edges from $P^{(\text{fused})}$ for removal. Let $\mathcal{M} \subset \mathcal{E}$ denote the set of masked edges, where $\mathcal{E}$ represents the complete edge set in the original fused graph. The model is then trained to reconstruct these masked edges based on the remaining observed graph structure.

For each masked edge $e = (u, v) \in \mathcal{M}$ with original weight w_e , we compute its reconstruction score using the learned embeddings. The reconstruction loss for a single masked edge is defined as:

$$\mathcal{L}_e = -\log \sigma(s_{uv}) \cdot w_e \quad (10)$$

where $\sigma(\cdot)$ is the sigmoid function, and $s_{uv} = \mathbf{z}_u^\top \mathbf{z}_v$ represents the predicted affinity between cells u and v .

To provide a proper training signal and prevent trivial solutions, we maintain the structural information from the observed edges. The complete reconstruction objective combines the masked edge recovery with the preservation of observed connectivity:

$$\mathcal{L}_{\text{recon}} = \sum_{e \in \mathcal{M}} \mathcal{L}_e + \alpha \cdot \sum_{e \in \mathcal{E}_{\text{obs}}} (\mathbf{z}_u^\top \mathbf{z}_v - w_e)^2 \quad (11)$$

where $\mathcal{E}_{\text{obs}}$ denotes the set of observed (non-masked) edges, and α balances the importance of edge reconstruction versus structure preservation.

To prevent overfitting and stabilize training, we incorporate L_2 regularization on the embedding parameters, leading to the full optimization objective:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda \cdot \|\mathbf{Z}\|_F^2 \quad (12)$$

where λ controls the regularization strength and $\|\cdot\|_F$ denotes the Frobenius norm [Peng et al. 2016].

The model’s ability to recover masked edges serves as a self-supervised signal that ensures the learned embeddings capture essential topological properties of the multimodal graph. Once trained, we extract the cell-specific embeddings $\mathbf{Z}_{\text{cells}} \in \mathbb{R}^{n \times d}$. The reconstructed cell–cell similarity network is then computed as:

$$\mathbf{W}_{\text{integrated}} = \mathbf{Z}_{\text{cells}} \cdot \mathbf{Z}_{\text{cells}}^\top \quad (13)$$

Therefore, this process, through the self-supervised graph reconstruction task, successfully transforms the high-dimensional and sparse multimodal similarity graph $P^{(\text{fused})}$ into a dense, low-dimensional, and semantically rich node feature matrix $\mathbf{Z}_{\text{cells}}$, laying a solid foundation for subsequent in-depth analysis. For cell clustering, we applied the Leiden algorithm [Traag et al. 2019] to the learned embeddings $\mathbf{Z}_{\text{cells}}$, which effectively partitions the graph structure of cells into biologically meaningful communities. The clustering was performed with a resolution parameter of 0.3 to obtain an appropriate level of granularity for distinguishing major cell types.

2.6 Model Evaluation

To evaluate the performance of various methods, we use three widely used evaluation metrics including Adjusted Rand Index (ARI) [Santos and Embrechts 2009], Normalized Mutual Information (NMI) [Estévez et al. 2009], and Adjusted Mutual Information (AMI)

[Romano et al. 2014]. ARI is used to measure the similarity between the model’s predicted categories and the true categories, defined as:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}} \quad (14)$$

The a_i and b_i denote the number of cells in the i -th real cluster and the i -th predicted cluster, respectively, and n_{ij} denotes the number of shared cells in the j -th real cluster and the j -th predicted cluster. NMI is used to assess the consistency between predicted and true categories. Let $T = t_1, t_2, \dots, t_n$ and $Y = y_1, y_2, \dots, y_n$ denote the true and predicted categories of cells, respectively, then define:

$$\text{NMI}(T, Y) = \frac{2\text{MI}(T, Y)}{H(T) + H(Y)} \quad (15)$$

The $H(\cdot)$ and $MI(\cdot)$ denote information entropy and mutual information, respectively. Considering that MI is higher if two clusters are of larger magnitude, regardless of whether there is more information to share, AMI adjusts the mutual information score, defined as:

$$\text{AMI}(T, Y) = \frac{\text{MI}(T, Y) - E(\text{MI}(T, Y))}{[H(T) + H(Y)]/2 - E(\text{MI}(T, Y))} \quad (16)$$

The $E(\cdot)$ denotes the mathematical expectation. The values of ARI and AMI are in the range of $[-1, 1]$, and the values of NMI are in the range of $[0, 1]$. The larger the values of these metrics, the better the performance of clustering.

3 Results

3.1 Performance comparison of multimodal integration methods

We benchmarked sxSNF against six representative multimodal integration frameworks on the PBMC-10k and SHARE-seq datasets (Tables 2 and 3). The results show that sxSNF achieves the highest performance across all three clustering metrics (ARI, NMI, AMI) on both datasets, demonstrating strong robustness and generalizability across tissue types and experimental protocols.

Performance on PBMC-10k. On the PBMC-10k dataset, sxSNF achieves an ARI of 0.5584, outperforming the next-best model, SIMBA (0.4854), by 15.0%. Similarly, sxSNF improves NMI and AMI by 6.4% relative to SIMBA. These gains arise from sxSNF’s fused similarity representation, which integrates complementary regulatory signals from both RNA expression and chromatin accessibility. In contrast, SIMBA learns joint embeddings by treating multi-omic relationships as entity relations, but lacks explicit neighborhood diffusion, causing reduced sensitivity to finer cell substructure. Spectral clustering, which relies solely on global similarity without modality alignment, performs worst across all metrics (ARI=0.0991), illustrating the necessity of modality coupling.

Comparison with scMIC and SNF+scMIC. The base model scMIC yields limited performance (ARI=0.1142), largely due to instability in graph construction when features are sparse or noisy. After incorporating SNF, the enhanced SNF+scMIC achieves a substantial improvement (ARI increases to 0.3523, +208.5% over scMIC).

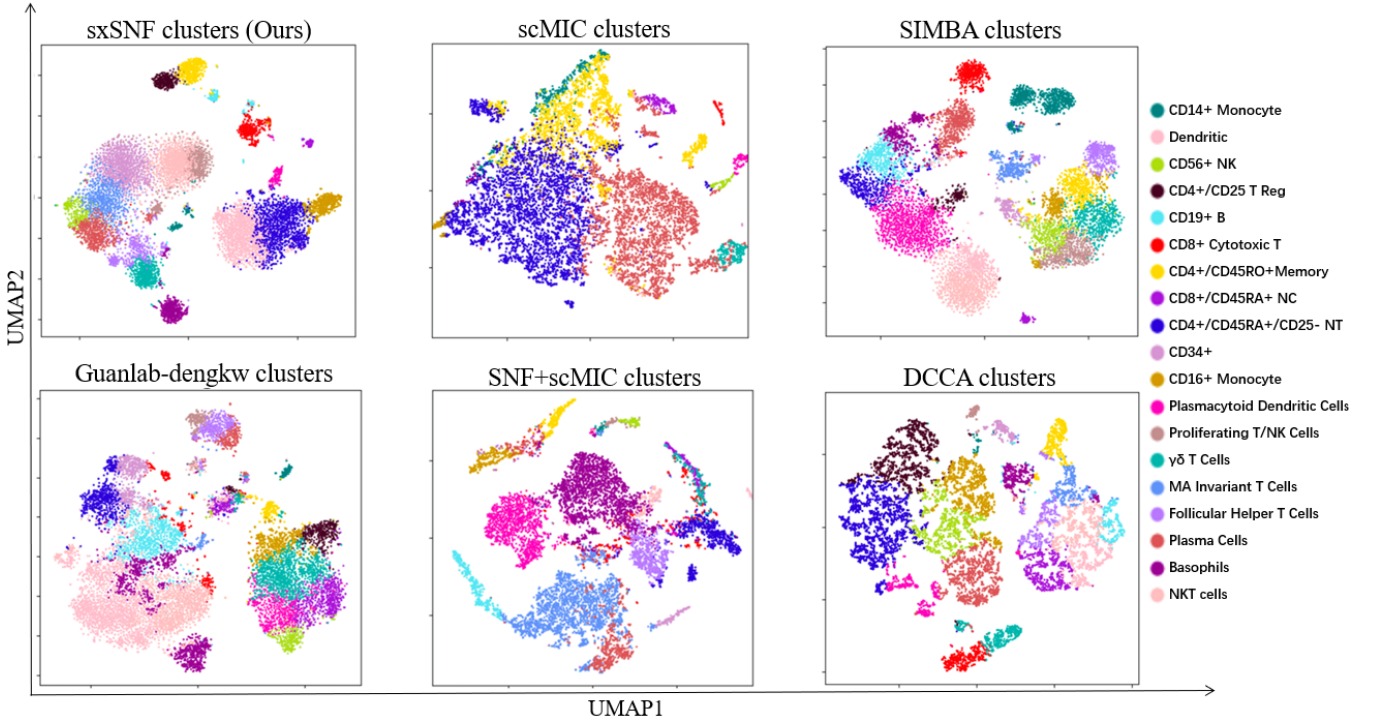


Figure 2: Visualization of clustering results comparing single-cell multimodal integration methods on the PBMC-10k dataset.

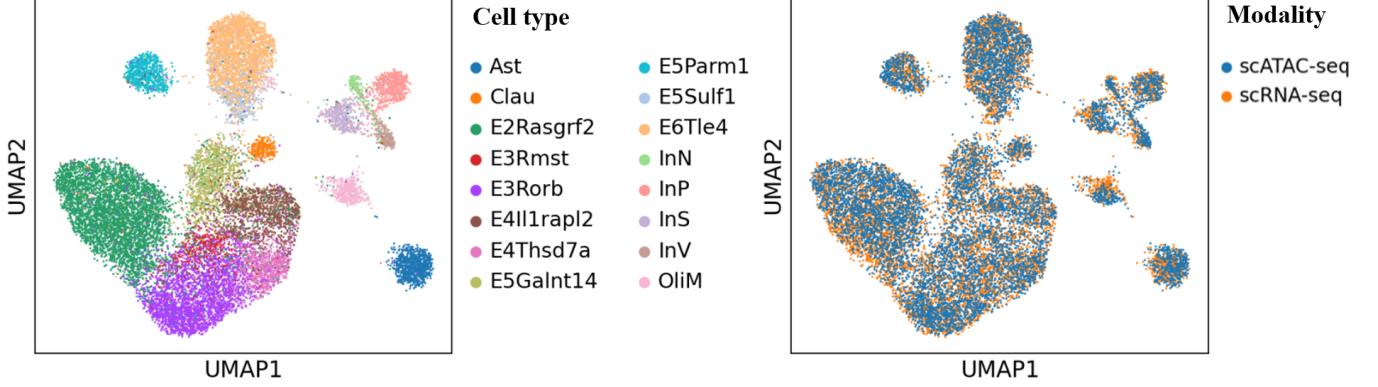


Figure 3: UMAP plots of sxSNF fused scRNA-seq and scATAC-seq embedding from the Chen-2019 Dataset.

This dramatic gain confirms the importance of cross-modal similarity diffusion, which corrects fragmented neighborhood structures. However, SNF+scMIC still does not reach sxSNF or SIMBA, because its embedding stage lacks non-linear feature refinement, retaining only aggregated pairwise similarity information.

Performance on SHARE-seq. A consistent trend is observed on the SHARE-seq dataset. sxSNF achieves ARI = 0.589 and NMI = 0.768, outperforming SIMBA by 15.0% ARI and 6.1% NMI, and surpassing DCCA by 49.5% ARI. The dataset contains greater regulatory heterogeneity from chromatin accessibility dynamics, which favors

models that explicitly propagate shared neighborhood information across modalities. Accordingly, scMIC again benefits from SNF (ARI increases from 0.153 to 0.421), confirming the generalizable reinforcement effect of SNF across datasets.

3.2 Synergistic benefits of SNF

As summarized in Table 4, the incorporation of SNF consistently enhanced the performance of baseline integration algorithms across ARI, NMI, and AMI metrics. The SNF+scMIC model exhibited the most pronounced improvement, with ARI increasing by 208.49%, and NMI and AMI more than doubling (+101.76% and +103.78%,

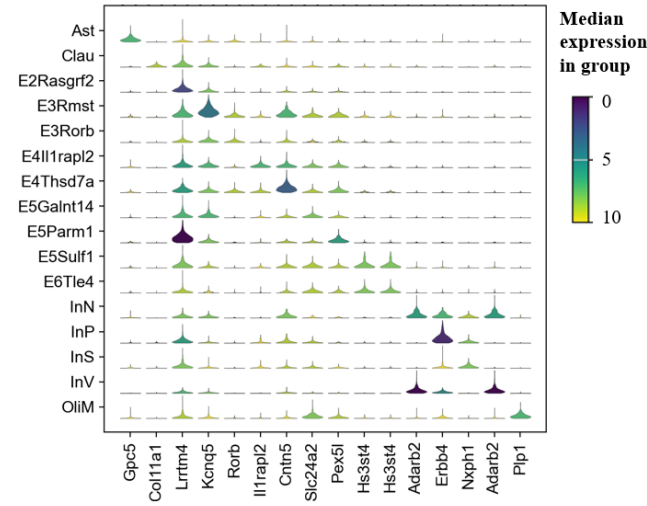


Figure 4: Landscape of cell-type-specific marker genes in the Chen-2019 dataset visualized by a stacked violin plot.

Table 4: Changes of clustering metrics after incorporating SNF

Methods	ARI	NMI	AMI
SNF+SIMBA	+15.04%	+6.36%	+6.42%
SNF+scMIC	+208.49%	+101.76%	+103.78%

Table 5: Effects of distance measures on the clustering results

Methodologies	Composition	ARI	NMI	AMI
sxSNF	cosine distance	0.5719	0.7582	0.7351
	chi-square distance	0.5443	0.7179	0.7206
SNF+scMIC	cosine distance	0.3811	0.6047	0.5928
	chi-square distance	0.3024	0.5223	0.5176

respectively). SNF+SIMBA also showed clear gains, with a 15.04% increase in ARI and approximately 6.4% improvements in NMI and AMI. These results indicate that SNF effectively leverages complementary omics information through iterative cross-modal diffusion, with synergistic effects being especially strong when applied to frameworks with greater latent capacity for refinement.

3.3 Effects of distance measures

We further examined the robustness of sxSNF by evaluating the influence of key components in the fusion framework. We investigated the role of distance measures in constructing the initial modality-specific cell graphs. As summarized in Table 5, cosine distance consistently outperformed chi-square distance across both sxSNF and SNF+scMIC, yielding higher ARI, NMI, and AMI scores. This advantage is likely due to the high sparsity and scale variability of single-cell molecular matrices, where cosine distance better captures directional consistency in expression patterns rather than absolute magnitude. Notably, sxSNF exhibited smaller performance

drops than SNF+scMIC when switching from cosine to chi-square distance, suggesting that the fused representation learned by sxSNF is more robust to shifts in initial graph construction. These analyses indicate that: cosine distance provides a more stable foundation for constructing cell-cell affinity graphs in multimodal fusion. Moreover, the relatively consistent performance of sxSNF across distance metrics highlights its resilience against variations in preprocessing and its suitability for diverse multimodal single-cell datasets.

3.4 Visualization and biological interpretability of integrated embeddings

To visually assess clustering quality and multimodal alignment, we applied uniform manifold approximation and projection (UMAP [McInnes et al. 2018]) to latent representations derived from each integration method. In Figure 2, embeddings are colored by ground-truth cell-type labels. The sxSNF method produced well-separated clusters with sharp boundaries, closely matching the underlying biological structure. Furthermore, a comparison of scMIC before and after SNF+scMIC integration revealed marked improvement in cluster cohesion and separation, corroborating the improved quantitative metrics in Table 2.

We further evaluated the joint embedding of scRNA-seq and scATAC-seq data from the Chen-2019 dataset (Figure 3). The left panel, colored by cell type, reveals 18 distinct clusters that correspond to known biological identities. The right panel, colored by assay modality (scATAC-seq in blue, scRNA-seq in orange), shows strong interleaving of both modalities within each cluster. This indicates successful removal of technique-specific bias while preserving discriminative cell-type structure in a unified latent space.

3.5 Marker gene expression supports cluster annotations

To validate the cellular identities obtained from the SNARE-seq dataset [Chen et al. 2019], we examined median gene expression patterns across annotated cortical populations. The observed distribution of canonical markers closely recapitulated established cortical taxonomy, indicating robust cluster assignment and high data integrity (see Figure 4).

Astrocytes (Ast) exhibited prominent expression of *Gpc5*, a glypican implicated in synapse development and previously reported as astrocyte-enriched [Bosworth et al. 2025]. Mature oligodendrocytes (OliM) showed strong expression of *Plp1*, a proteolipid essential for myelin formation and a well-established lineage marker [Gudz et al. 2002]. These glial signatures are consistent with recent cellular atlas and functional studies [Tasic et al. 2016].

Among excitatory neurons, transcriptomic profiles resolved into layer-specific subtypes. Cells annotated as E3 Rorb expressed high levels of *Rorb*, a hallmark of layer 4/4–5 intratelencephalic neurons [Clark et al. 2020]. Similarly, E4 Il1rapl2 populations were enriched for *Il1rapl2*, an upper-layer excitatory marker reported in recent single-cell cortical atlases [Franjic et al. 2022]. These patterns align with known laminar organization of cortical projection neurons [Tasic et al. 2016].

Inhibitory interneuron groups (InN, InP, InS, InV) displayed elevated *Adarb2*, a defining feature of CGE-derived interneuron

lineages [Hodge et al. 2019]. Additional neuronal markers demonstrated coherent cell-class specificity: *Kcnq5*, a potassium channel subunit contributing to neuronal excitability, was enriched broadly across neuronal populations but absent in glia [Fidzinski et al. 2015]. Synaptic adhesion genes such as *Nxph1* were preferentially expressed in neuronal clusters, including interneuron subsets [Born et al. 2014]. These trends are consistent with electrophysiological and single-cell transcriptomic evidence [Tasic et al. 2016].

4 Discussion and conclusion

In this study, we present sxSNF, a multimodal single-cell integration framework that combines Similarity Network Fusion (SNF) with deep graph learning to construct a unified and biologically meaningful cell–cell similarity representation. By iteratively diffusing neighborhood structures across modalities and subsequently learning a low-dimensional embedding on the fused graph, sxSNF effectively preserves complementary modalities signals while mitigating noise and batch-induced modality bias. Across multiple benchmark datasets, sxSNF consistently outperformed state-of-the-art integration approaches in clustering accuracy and biological interpretability, demonstrating its capability to recover fine-grained cellular heterogeneity.

Our parameter sensitivity analysis further highlighted two important characteristics of the framework. First, incorporating SNF as a structural diffusion layer substantially enhanced the performance of existing integration models, particularly in architectures where the initial graph representation is weak (e.g., scMIC). This suggests that SNF functions as a robust and generalizable enhancement mechanism, capable of strengthening cell–cell affinity signals before downstream representation learning. Second, cosine-based distance proved more effective than chi-square distance for constructing local affinity graphs, likely due to the high sparsity and compositional noise inherent to single-cell expression and accessibility data. Notably, sxSNF maintained stable performance regardless of similarity metric selection, reflecting the resilience of its fused graph representation.

Beyond improved clustering performance, sxSNF generated cell embeddings that aligned strongly with well-established lineage hierarchies and marker gene programs, as demonstrated in Figure 4. This biological coherence indicates that sxSNF not only integrates modalities at the computational level, but also preserves meaningful molecular identity and regulatory context across diverse cell types.

The novelty of sxSNF lies in its ability to combine SNF’s iterative fusion of data with the powerful learning capabilities of GNNs. At its core is Similarity Network Fusion (SNF), which iteratively combine similarity matrices from different data sources to capture both local and global relationships between cells. This integration allows for more accurate, biologically meaningful clustering of cells, preserving cellular heterogeneity while mitigating the effects of noise and batch differences. In addition to single-cell data, this technique has shown success in other domains, such as drug discovery, where it is used to integrate diverse biological data types to predict drug-target interactions. This success outside of single-cell research suggests that sxSNF could have broad applicability in other fields that require multi-modal data integration.

Looking forward, the sxSNF framework provides several avenues for extension. Its modular design allows straightforward incorporation of additional omics layers such as proteomics or spatial transcriptomics. Moreover, the fused graph representation can serve as a foundation for advanced downstream tasks, including trajectory inference, perturbation response modeling, and cell–cell communication analysis. Future work will explore adaptive fusion weights and end-to-end optimization strategies, enabling dynamic coupling between fusion and embedding stages.

In conclusion, sxSNF offers a robust, scalable, and biologically grounded solution for multimodal single-cell data integration. By combining the structural refinement power of SNF with the representational capacity of deep graph learning, sxSNF establishes a general framework for uncovering cellular identities and regulatory relationships in complex biological systems.

Acknowledgments

This study was funded by the National Science Foundation of China (12571529) and Guangdong Basic and Applied Basic Research Foundation (2024A1515-010699) to LCX.

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